HealthConnect: Week 6 KPI Validation (Data Analytics Track)

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Task: Week 6 brief, Data Analytics track, item 5: "Validate the most important KPIs developed in Week 5."

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1. What to expect this week

Week 5 calculated each KPI once, as a single point estimate on the full dataset. That tells you what the rate is, not how much that number would move if the sample had come out slightly differently. A point estimate with no sense of its own stability looks identical to a fragile number that just happened to land somewhere.

Method: bootstrap resampling. For each KPI, resample the underlying rows 1,000 times with replacement (same size as the original group), then report the resulting 95% confidence interval next to the Week 5 point estimate. Full working and outputs are in `notebooks/HealthConnect_Week6_KPI_Validation.ipynb`.

A KPI whose bands have confidence intervals that don't overlap is robust; the pattern would likely reappear with a different sample. A KPI whose bands overlap isn't necessarily wrong, but the point estimate shouldn't be presented with the same confidence as a robust one.

2. KPI 1: No-Show Rate by Prior No-Show Bracket

Bracket	n	Point estimate	95% CI
0	2,921	43.5%	[41.6%, 45.4%]
1	1,548	53.5%	[51.0%, 56.1%]
2	438	59.4%	[54.6%, 63.5%]
3+	93	68.8%	[58.1%, 78.5%]

Validated, with one caveat carried over from Week 5 and now measured precisely. Brackets 0, 1, and 2 separate cleanly with no overlap between adjacent pairs. Bracket "3+" (93 rows) has a wide interval, 58.1% to 78.5%, that overlaps with bracket 2's 54.6% to 63.5%. Week 5 flagged this bracket's small size as a concern; the bootstrap makes it exact. We cannot statistically tell "3+ prior no-shows" apart from "2 prior no-shows" at this sample size. The overall rising-risk pattern holds. The exact number for the "3+" tail doesn't.

3. KPI 2: No-Show Rate by Booking Lead-Time Band

Band	n	Point estimate	95% CI
0-7 days	640	27.8%	[24.4%, 31.1%]
8-14 days	602	33.6%	[29.7%, 37.4%]
15-30 days	1,333	43.2%	[40.7%, 45.9%]
31-45 days	1,258	53.8%	[51.0%, 56.6%]
46-60 days	1,167	67.7%	[65.2%, 70.2%]

Fully validated. This is the strongest KPI in the set. Every band separates cleanly from every non-adjacent band, and even neighboring bands barely touch (0-7 vs 8-14: 31.1% vs 29.7%). The climb from 27.8% to 67.7% is a real, stable pattern, not an artefact of how the data happened to split. Three separate checks now agree on this: the Week 5 point estimates, this bootstrap, and the Data Science track's cross-validated model, where `booking_lead_days` came out as the single most important predictive feature by a wide margin (permutation importance 0.097-0.105, five to ten times any other feature).

4. KPI 3: No-Show Rate by Distance Band

Band	n	Point estimate	95% CI
<5 km	1,156	46.5%	[43.5%, 49.4%]
5-10 km	1,692	46.5%	[44.1%, 48.8%]
10-15 km	1,132	48.5%	[45.6%, 51.6%]
15+ km	930	54.1%	[51.0%, 57.4%]

This KPI changed this week. The three middle bands, under 5km, 5-10km, and 10-15km, have confidence intervals that overlap almost completely: the data can't reliably tell these groups apart. Only the two extremes separate at all, and even then just barely, 49.4% against 51.0%, a gap of 1.6 points.

Nothing was wrong with the Week 5 calculation. The point estimates are correct. What this check catches is that a real gradient in the point estimates can still be too noisy to call a "cliff" with any confidence.

The Data Science track found the same thing independently, working from the same processed dataset but a completely different method: permutation importance on a trained classifier, not confidence intervals on a rate. `distance_to_clinic_km` and a `distance_15km_plus` threshold flag both came out with negligible predictive value next to lead time and prior no-shows. Two unrelated methods landing on the same downgrade is stronger evidence than either one alone.

Distance-based intervention, transport support or a telehealth option, should now be framed as a hypothesis worth a small, measured pilot, not the confirmed lever Week 5 implied.

5. KPI 4: Wasted Slot Rate by Reminder Status

Group	n	Point estimate	95% CI
Reminder sent	3,634	47.4%	[45.8%, 49.0%]
No reminder	1,366	51.4%	[48.7%, 54.2%]
Overall	5,000	48.5%	[47.0%, 49.9%]

Confirmed weak, now precisely. The two groups' intervals nearly touch, 49.0% against 48.7%, a 0.3-point gap, which means the roughly 4-point reminder effect sits right at the edge of what this sample size can distinguish from noise. That matches the Data Science model's near-zero permutation importance for reminder status. Week 5 already called this "worth keeping, but nowhere near enough on their own." That framing turns out to have been right all along.

6. KPI 5: Cancellation-to-No-Show Ratio

Group	n	Point estimate	95% CI
Overall	5,000	0.109	[0.095, 0.122]
0-7 days	640	0.202	[0.138, 0.286]
8-14 days	602	0.139	[0.088, 0.197]
15-30 days	1,333	0.118	[0.090, 0.148]
31-45 days	1,258	0.105	[0.079, 0.132]
46-60 days	1,167	0.076	[0.057, 0.097]

Validated. The two ends of the range, 0-7 days at [0.138, 0.286] and 46-60 days at [0.057, 0.097], don't overlap at all. Longer lead time means proportionally more silent no-shows relative to proper cancellations, and that pattern is real, not sampling noise.

7. Summary: what changed and what didn't

KPI	Week 5 status	Week 6 validation result
1: Prior no-show bracket	Point estimate only	Robust for brackets 0-2; "3+" confirmed fragile (already flagged, now quantified)
2: Lead-time band	Point estimate only	Fully validated, strongest KPI, confirmed by three independent methods
3: Distance band	Point estimate, framed as a "cliff"	Downgraded: middle bands statistically indistinguishable, corroborated by the DS model

KPI	Week 5 status	Week 6 validation result
4: Wasted slot rate / reminders	Point estimate, framed as "weak but real"	Confirmed at the same confidence level, no change needed
5: Cancellation-to-no-show ratio	Point estimate only	Validated, the trend is real and stable

KPI 3 is the one that moves the Week 6 narrative. Every recommendation and dashboard element built on the "15km cliff" should carry a lower confidence label from here on, and the distance-based recommendation should be reframed as a pilot rather than a confirmed lever.